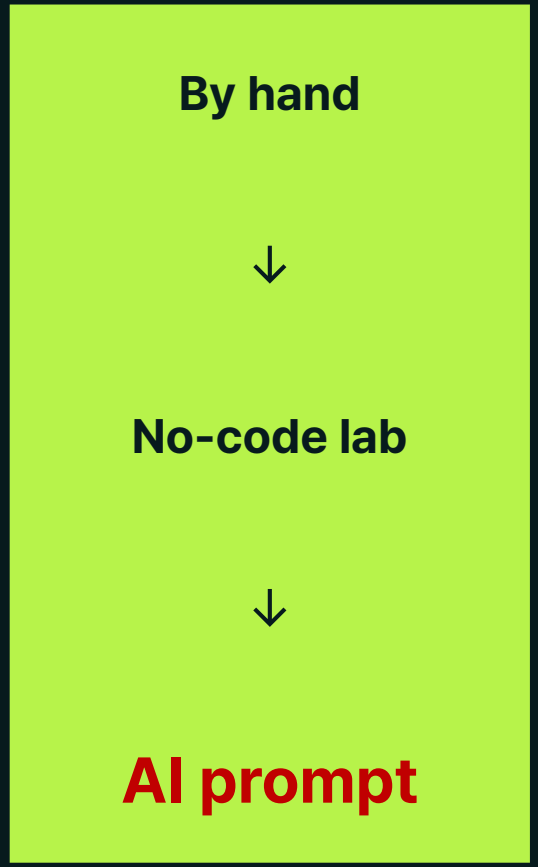
NLP · LLM · RAG Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 12 Workbook

Cosine similarity compares direction, not length

Three vectors versus reference (1, 0)

Step 1 · Same

$$\cos\big((1,0),(2,0)\big) = \frac{2}{1 \times 2} = 1$$

Step 2 · Right angle

$$\cos\big((1,0),(0,1)\big) = \frac{0}{1 \times 1} = 0$$

Step 3 · 45 degrees

$$\cos\big((1,0),(1,1)\big) = \frac{1}{1 \times \sqrt{2}} \approx 0.707$$

Zero vectors need separate handling (denominator 0)

Softmax maps 2, 1, −1 to 70.5%, 25.9%, 3.5%

‘Stocks yesterday ___’ — rose 2, fell 1, banana −1, T=1

Step 1 · Source

$$P_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}, \qquad z = \text{candidate score}, \quad T = \text{temperature}$$

Step 2 · Exponentiate, then sum

$$e^{2} = 7.389, \quad e^{1} = 2.718, \quad e^{-1} = 0.368, \quad \text{sum} = 10.475$$

Step 3 · Normalize

$$\frac{7.389}{10.475} \approx 70.5\%, \quad \frac{2.718}{10.475} \approx 25.9\%, \quad \frac{0.368}{10.475} \approx 3.5\%$$

The likeliest word need not be true

Answering from three documents only shows how RAG works

Paste mock documents A, B, C into an AI chat

1. Where

Open an AI chat.

2. Input

Paste A (30-day refund), B (rate 3%), C (sales ₩10bn).

3. Output

‘Refund period?’ gets 30 days, citing A.

4. Check

Check it admits no evidence on ‘Overseas buys?’

Prompt for the AI chat

Answer only from the three documents below. Pick the relevant document IDs, quote the evidence briefly, then answer. Do not guess missing conditions; say ‘cannot be verified’. Check numbers in the question against the documents. [Paste documents and question]

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.